Akhil Tabjula

Senior Data Engineer / AI Developer
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SUMMARY

Senior Data Engineer with 10+ years of experience across the full data engineering lifecycle. I have worked through several technology cycles — from Hive and MapReduce to Spark and Databricks, from batch ETL to real-time Kafka pipelines, and more recently into agentic AI frameworks. Across all of it, the focus has been the same: building data infrastructure that is reliable, well-governed, and close enough to decisions to actually influence them.

SKILLS

Cloud & Platforms: Azure (Data Factory, Databricks, Functions, DevOps, SQL Database, Data Lake), Snowflake, AWS
Big Data & ETL: Apache Spark, PySpark, Kafka, Airflow, Hive, MapReduce, HDFS, Talend, IBM DataStage
AI & LLM: LangGraph, LangChain, Agentic AI, Stateful Agents, RAG, MongoDB Atlas Vector Search, Embeddings, MCP, LLM Integration
Languages: Python, SQL, Scala, C#
Data Modeling & BI: dbt, Snowpipe, Streams, Tasks, CDC, Power BI, Tableau
DevOps & Tools: Git, GitHub, Jenkins, Nexus, Azure Pipelines, CI/CD, TAC, CARTO (H3, Quad Keys), REST APIs

PROFESSIONAL EXPERIENCE

Senior Data Engineer | Coca-Cola Southwest Beverages **Oct 2023 – Present**

- Built an agentic API orchestrator behind a web chat interface using LangGraph and stateful agents that tells users which internal API, endpoint, scope, and contract fits their use case.
- Developed a backend indexing pipeline that scans all organization GitHub repositories, extracts API metadata (endpoints, resources, scopes, contracts), and generates embeddings into a central API registry in MongoDB Atlas for vector search retrieval (RAG).
- Designed an execution agent that identifies the target API, gathers missing payload parameters through multi-turn clarifying questions, invokes the API, and transforms raw responses into human-readable answers.
- Architected a real-time recommendation engine (Suggested Order) combining historical sales with ML models to predict customer potential and drive order recommendations.
- Led trade promotion analytics, building predictive allocation algorithms that improved promotion ROI by 1.8% across key accounts.
- Built a Spark optimization framework for complex transformations and predictive analytics, speeding delivery across 3 concurrent projects.
- Built Azure Function Apps and REST APIs connecting Power BI, Databricks, Dataverse, Data Lake, and Azure SQL for KPI-aligned data flow.
- Implemented agentic automation for multi-level predictive analysis with token-efficient LLM integration; integrated PredictHQ event data for real-time order prediction.
- Applied geospatial transformations via CARTO workflows with H3 indexing and Quad Keys for location intelligence.

Senior Data Engineer | Victoria's Secret **Jun 2021 – Sep 2023**

- Redesigned data modeling using dbt, improving data accuracy by 30% and streamlining downstream reporting across the organization.
- Led full migration of Teradata batch jobs to Snowflake, delivering performance gains and cost savings; migrated IBM DataStage ETL processes to Talend.
- Built real-time Snowflake pipelines for live sales transactions using Snowpipe, Streams, and Tasks, with error monitors and auto-reprocessing.
- Implemented CDC in Talend, designed JSON-to-Variant ingestion via external stages and COPY commands, and established CI/CD using Git, Jenkins, Nexus, and TAC.
- Enforced data governance through Talend Data Catalog and applied compliance standards across the full Azure data lifecycle.

Data Engineer | Inspire Brands **Jan 2021 – May 2021**

- Orchestrated Azure Data Factory pipelines unifying sales data from transaction databases and online platforms; implemented Azure Pipelines CI/CD for PySpark and Snowflake deployments.
- Built PySpark frameworks in Databricks for distributed processing, enabling sales forecasting, customer segmentation, and product affinity analysis at scale.
- Architected Snowflake schemas for retail analytics and integrated Power BI dashboards for real-time revenue, margin, and trend visibility.

Project Engineer | Wipro Technologies **Aug 2014 – Aug 2019**

- Designed SQL Server objects (tables, views, stored procedures, indexes) for staging and data lake environments; managed source-to-target mapping for all facts and dimensions.
- Ingested data from diverse sources into HDFS using Hive and MapReduce; built Azure Data Factory orchestration workflows.
- Resolved on-call Spark production failures within SLA, mentored junior engineers, and built Tableau dashboards for data science.
- Developed features for the Dynamics 365 Resource Scheduling Optimization (RSO) module using C#, SAAPI, and Azure.

EDUCATION

Master of Science in Data Science | University of North Texas **Aug 2019 – May 2021**

Bachelor of Technology | KL University **2010 – 2014**